

ACCHLT: An Adaptive Cross-Coupled Hénon–Logistic–Tent Chaotic Map for Plaintext-Aware Two-Round Image Encryption

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ABSTRACT

Image encryption schemes built on low-dimensional chaotic maps often suffer from small key space, short periodic windows, and predictable trajectories once the underlying map is identified. This paper proposes ACCHLT, a novel three-dimensional discrete chaotic map that cross-couples a logistic-type term, a Hénon-type quadratic term, and a conditional tent-type term through three adaptive coupling parameters, with the growth-rate parameters of the logistic and tent components themselves varying at every iteration as functions of the other two state variables. Bifurcation analysis and the Lyapunov exponent spectrum show that the map exhibits robust chaotic behaviour, with a maximum Lyapunov exponent of 0.654 sustained across the explored parameter range, in contrast to the narrow chaotic windows of the classical Hénon map used as a baseline. Bitstreams derived from ACCHLT pass all 14 applicable NIST SP 800-22 statistical tests and all 15 statistics of the TestU01 SmallCrush battery. Using ACCHLT, a two-round image encryption scheme is developed in which the initial conditions of round one are derived from a 256-bit key together with an HMAC-SHA256 digest of the plaintext image, and the initial conditions of round two are further coupled to the intermediate ciphertext, giving the scheme resistance to chosen-plaintext attacks beyond what a purely key-dependent design provides. Each round applies chaos-driven pixel permutation, a dynamically generated substitution box, and feedback-chained diffusion. Experiments on 38 standard USC-SIPI test images show a mean NPCR of 99.6066%, mean UACI of 33.4670%, mean information entropy of 7.9024 bits (all within 0.01 of the theoretical ideal), adjacent-pixel correlation reduced to near zero in all directions, and exactly lossless decryption ($MSE = 0$, $PSNR = \infty$) in every trial, indicating that ACCHLT provides a competitive and practically efficient basis for secure image transmission.

1. INTRODUCTION

1.1. Background and Motivation

The volume of digital imagery exchanged over public and semi-trusted networks — in surveillance, telemedicine, defence reconnaissance, remote sensing, and personal communication — has made image confidentiality a central concern in applied cryptography (1; 2). Unlike generic text or numeric data, digital images possess properties that make them poorly suited to conventional encryption: very high inter-pixel redundancy, strong local spatial correlation between neighbouring pixels, large file sizes that must be processed efficiently, and, in many applications, a requirement that the encryption process itself be fast enough to support near-real-time capture and transmission. Classical block ciphers such as the Advanced Encryption Standard (AES) (4) were designed and standardised for generic byte streams under exactly these assumptions violated by images, and applied directly to image data in a naïve mode of operation, AES can leave detectable residual structure in the cipher output — visible edges or texture patterns surviving encryption — while also being comparatively expensive to run at image-sized payloads in real time (1).

The practical need for dedicated image ciphers spans several distinct domains, each with its own confidentiality requirements. In telemedicine and hospital picture-archiving systems, medical images (X-rays, MRI, CT, ultrasound) are subject to patient-privacy regulation and must remain confidential both at rest and in transit (3). In defence, surveillance, and remote-sensing applications, reconnaissance and satellite imagery must be protected against interception on public or adversarial networks, often under tight latency constraints (1). In consumer and cloud contexts, the sheer volume of personal photographs uploaded to third-party storage has made client-side or transport-layer image encryption an increasingly common requirement rather than a specialised one (21; 22). These differing requirements share a common technical core — the need for a cipher that destroys the statistical structure of the plaintext image efficiently and reproducibly — which is the problem addressed by chaos-based image encryption in general and by this paper in particular.

ORCID(s):

1.2. Threat Model and Design Requirements

A cryptographically strong image cipher must simultaneously defeat several broad classes of attack: statistical attacks that exploit non-uniform histograms or residual pixel correlation; differential attacks that exploit predictable relationships between small changes in the plaintext and the resulting changes in the ciphertext; brute-force attacks against an insufficiently large key space; and, increasingly, targeted cryptanalytic attacks that exploit implementation-specific weaknesses in a scheme's key-dependence rather than attacking the underlying map directly (17; 18). This last point is often under-appreciated: a chaotic map can be shown to be highly chaotic in isolation and yet the resulting cipher can still be broken if the mapping from the secret key (and, ideally, the plaintext) to the map's initial conditions is not carefully engineered (18). This paper treats key-dependence, and more specifically plaintext- and ciphertext-dependence, as a first-class design requirement rather than an afterthought.

Chaos-based cryptography has become a dominant approach to meeting these requirements, since chaotic systems naturally provide the two properties Shannon identified as essential to a strong cipher — confusion and diffusion (8) — through extreme sensitivity to initial conditions, ergodicity, and pseudo-random long-term behaviour that is nonetheless fully deterministic and reproducible given the correct seed. Early designs relied on one-dimensional maps such as the logistic, sine, and tent maps because of their mathematical simplicity and cheap per-iteration cost, but these maps are well documented to suffer from narrow chaotic ranges, periodic windows, and blank/stable windows that reduce the effective key space and make trajectory prediction considerably easier once the specific map is known to an attacker (1; 3; 21; 22). This has motivated a sustained shift, over roughly the last decade, toward two- and higher-dimensional chaotic maps — such as the Hénon map and its many hybrid variants — which offer wider chaotic ranges and stronger diffusion at a moderate additional computational cost (1; 3), and, more recently, toward three-dimensional and n-dimensional constructions that allow more of the Lyapunov spectrum itself to be shaped or controlled by design (11).

1.3. Proposed Approach

This paper continues that line of work by proposing a new three-dimensional chaotic map, referred to as ACCHLT (Adaptive Cross-Coupled Hénon–Logistic–Tent map), and a two-round image encryption scheme built on it. Unlike a simple concatenation of three independent 1D maps — which would merely inherit the periodic-window weaknesses of each component — the three state variables of ACCHLT are mutually coupled at every iteration: the logistic-type growth rate and the tent-type growth rate are themselves functions of the other state variables rather than fixed constants, and a bifurcated (tent-like) update rule is applied to the third variable. This cross-coupling is shown in Section 4 to suppress the periodic windows that would otherwise interrupt the chaotic range of any one component map taken alone.

In addition, and directly motivated by the key-dependence concerns raised above, the proposed encryption scheme ties the initial conditions of its two rounds not only to a 256-bit secret key but also to the plaintext image itself (via an HMAC-SHA256 digest of a pixel sample) and to the intermediate ciphertext produced by the first round, so that even a single-bit change anywhere in the plaintext propagates into a completely different keystream for both rounds — a property verified quantitatively in Section 6.4. The problem addressed in this paper can therefore be stated concisely: design a chaotic map whose chaotic behaviour is robust across a wide parameter range rather than confined to narrow windows, and use it to build an image cipher whose effective keystream depends jointly on the secret key, the plaintext, and the evolving ciphertext, so that the resulting scheme resists differential, statistical, and brute-force attacks while remaining efficient enough for practical use.

1.4. Contributions

The main contributions of this work are summarised as follows:

- A novel adaptive, cross-coupled 3D chaotic map (ACCHLT) is proposed in which the logistic and tent-type growth-rate parameters vary dynamically with the other state variables, rather than remaining fixed constants as in the classical Hénon or logistic maps.
- The chaotic behaviour of ACCHLT is validated using bifurcation diagrams and the full three-exponent Lyapunov spectrum, and its output bitstreams are validated against the NIST SP 800-22 and TestU01 SmallCrush statistical test suites.
- A plaintext- and ciphertext-aware two-round image encryption scheme is designed on top of ACCHLT, combining chaotic permutation, dynamically generated S-boxes, and feedback-chained diffusion, with round-two initial conditions coupled to round-one output for additional resistance to chosen-plaintext attacks.

- The scheme is evaluated on 38 standard USC-SIPI test images using histogram uniformity, adjacent-pixel correlation, information entropy, NPCR/UACI differential-attack resistance, key sensitivity, and lossless-recovery (MSE/PSNR) analysis, and is compared against several recently published chaos-based image encryption schemes.

1.5. Paper Organization

The remainder of this paper is organised as follows. Section 2 surveys the broader landscape of chaos-based image encryption research. Section 3 reviews the three works most closely related to the present scheme in detail. Section 4 introduces the ACCHLT map and validates its chaotic properties. Section 5 describes the proposed encryption and decryption scheme. Section 6 presents the experimental results and comparative analysis. Section 7 concludes the paper.

2. LITERATURE SURVEY

Chaos-based image encryption is now a mature research area, and two recent survey papers give a useful map of the field. Zia et al. (21) survey chaotic image encryption techniques across the spatial, transform, and spatiotemporal domains, classifying schemes by the domain in which permutation and diffusion are applied. Fang et al. (22) provide a broader survey of the whole chaotic-system image encryption literature, tracing the field from simple 1D-map schemes through DNA-coding hybrids, transform-domain and compressive-sensing schemes, to hyperchaotic and learning-assisted designs, and highlighting the classification of chaotic systems used throughout this paper's own related work discussion.

Within low-dimensional constructions, several works have continued to refine 1D chaotic maps specifically to avoid the periodic-window problem discussed in Section 1. Talhaoui et al. (23) proposed a new one-dimensional chaotic map used in a permutation-less image encryption scheme, folding permutation and diffusion into a single pass to reduce computation. Mansouri and Wang (25) combined a shuffled Arnold map with multiple-value pixel manipulations for a lightweight block-based cipher. These works, together with the Sine-Tangent map of Dua and Bhogal reviewed in detail in Section 3, illustrate that even 1D maps remain competitive when carefully engineered, though they typically still require an external mechanism (larger keys, DNA coding, or coupling with a second map) to reach the statistical quality of higher-dimensional constructions.

DNA-coding hybrids form a second major stream of work. Wang et al. (24) combined Fisher–Yates pixel scrambling with a DNA-subsequence diffusion operation, part of a broader family of schemes in which chaotic sequences drive the choice of DNA encoding rule and DNA-domain arithmetic (addition, XOR, complement) replaces or supplements conventional pixel-domain diffusion. Such schemes can achieve very high nonlinearity in the substitution stage, but Fang et al. (22) note that the computational overhead of DNA encoding/decoding is often significant relative to the security gained, which is one reason the present work retains conventional pixel-domain S-box substitution rather than adopting a DNA layer.

Plaintext- and key-dependence has emerged as an important design axis distinct from the choice of chaotic map itself. Zhang et al. (19) proposed a plaintext-related scheme in which a perceptron-like network derives part of the keystream from the plaintext, so that different images produce structurally different diffusion even under an identical key. Wu et al. (27) similarly designed a plaintext-related dynamic-key chaotic scheme, updating the effective key during encryption as a function of the data already processed. Both works motivate the HMAC-SHA256 plaintext-coupling mechanism adopted in Section 5.2 of the present paper, though the present scheme extends the idea further by also coupling the second encryption round to the first round's ciphertext output, which neither of these two related designs does.

A smaller but growing body of work uses learning-based or number-theoretic mechanisms to strengthen key generation itself, complementing rather than replacing the chaotic map. Ding et al. (26) proposed DeepKeyGen, a deep-learning-based stream-cipher generator for medical image encryption, learning a key-stream generator directly from training data rather than deriving it analytically from a chaotic map. Al-Hyari et al. (28) instead strengthened chaotic key generation using the Collatz conjecture to expand the effective key space available to a chaotic map-based cipher. Neither direction is adopted directly in this paper, but both illustrate that key-space and key-generation engineering is now treated as being as important as the choice of chaotic map, a view reflected in the deliberately conservative key-space accounting given in Section 6.7 and informed by the keyspace-analysis caution of Abba et al. (17) and the concrete cryptanalytic break reported by Noshadian et al. (18).

Finally, and closest in spirit to the present contribution, higher-dimensional chaotic map design has increasingly focused on controlling or shaping the Lyapunov spectrum directly rather than only demonstrating that some parameter range is chaotic; this trend, and the specific works of Cheng et al. (11), Qiu et al. (12), Zhang et al. (13), Kocak et al. (14), Lai et al. (15), and Zhao et al. (16), is discussed further in Section 3 alongside the three schemes to which the proposed ACCHLT map and cipher are most directly compared.

3. RELATED WORK

Bayesh et al. (1) proposed a dual-layer framework that strengthens AES-128 with chaotic components — a two-dimensional Hénon map for generating a dynamic, round-wise substitution box, and a logistic map for a dynamic ShiftRows pattern — combined with a hybrid CBC/ECB block-chaining mode. To address secure key delivery, the same work introduced a steganographic QR-code channel in which a static key and an ElGamal-encrypted dynamic session key are embedded using least-significant-bit (LSB) steganography, so that the two keys can travel openly alongside the encrypted image without exposing either key directly. Reported results include near-ideal entropy (≈ 7.997 bits/pixel), NPCR above 99.6%, and UACI in the 33–34% range, together with robustness to cropping, data loss, and several noise models. The QR-based key channel, however, was shown to be fragile under LSB perturbation, illustrating a trade-off between covert key delivery and decoding robustness that the present work sidesteps by deriving all chaotic parameters directly and deterministically from a shared secret key rather than transmitting them visually.

Dua and Bhogal (2) focused on the design of a new one-dimensional chaotic map rather than a hybrid AES construction. Their Sine-Tangent Chaotic (STC) map combines sine and tangent transformations with a large exponent term, and was shown via bifurcation diagrams, Lyapunov exponent, and Shannon entropy analysis to have a wider and more consistently chaotic range than the classical logistic, sine, and tent maps. The STC map was then used, together with a 384-bit XOR/permutation-based shared key, to build a lightweight medical image cipher. On medical image datasets, the scheme achieved an information entropy of 7.99977, NPCR of 99.66023%, and UACI of 33.80065%, and passed a full suite of NIST SP 800-22 tests. This motivated the present paper's decision to validate the proposed 3D map against the same battery of statistical tests rather than relying on bifurcation and Lyapunov analysis alone.

Sharma et al. (3) proposed a two-dimensional Hénon–Sine map (2D-HSM), obtained by nesting a modified Hénon map inside a sine function, and showed — through trajectory plots, bifurcation diagrams, the full two-exponent Lyapunov spectrum, and Kolmogorov entropy — that the resulting map is hyperchaotic (both Lyapunov exponents positive) across an extensive parameter range. A two-round image encryption algorithm built on 2D-HSM, seeded by a 256-bit key and using a randomised pixel border, achieved near-zero adjacent-pixel correlation and NPCR/UACI scores closely matching the theoretical ideal (99.6094%/33.4635%) across a twenty-eight-image benchmark set. This paper follows a broadly similar validation methodology but extends the underlying map to three coupled, adaptively parameterised dimensions, and extends the key-dependence of the cipher to also depend on the plaintext and on the first round's ciphertext output.

Beyond these three works, three-dimensional and higher-dimensional chaotic maps have continued to attract attention specifically because they allow more of the Lyapunov spectrum to be shaped by design. Cheng et al. (11) recently proposed a 3D Improved Logistic-Embedded Sine–Cosine Map with a dynamic, state-dependent exponential term that yields controllable Lyapunov exponents, applied to multi-image holographic encryption. Coupled two-dimensional constructions have also proven effective for colour imagery: Qiu et al. (12) combined a hyperchaotic map with Rubik's-cube-style scrambling, Zhang et al. (13) proposed a 2D sine–cosine coupling map, and Zhao et al. (16) introduced a further two-dimensional chaotic system for colour image encryption. Metaheuristic and modular constructions have also been explored, including a PSO-tuned modular integrated logistic-exponential map (14) and a pixel-split scheme built on a 2D Salomon map (15). Across these works, the general trend mirrors the motivation of this paper: cross-coupling multiple simple chaotic components tends to produce wider, more controllable chaotic behaviour than any single component map alone.

4. THE PROPOSED ACCHLT CHAOTIC MAP

4.1. Preliminaries

The classical logistic map is a one-dimensional map defined by $x_{n+1} = r \cdot x_n \cdot (1 - x_n)$, which is chaotic for $r \in (3.57, 4]$ but suffers from periodic windows within this range. The Hénon map (5) is a two-dimensional quadratic map defined by $x_{n+1} = 1 - a \cdot x_n^2 + y_n$, $y_{n+1} = b \cdot x_n$, and is chaotic for the classical parameter choice $a = 1.4$, $b = 0.3$,

Table 1

Positioning of the proposed scheme relative to reviewed related work.

Scheme	Map dimensionality	Plaintext/ciphertext coupling	External randomness suite
Bayesh et al. (1)	2D (Hénon + logistic)	No (key-only)	No
Dua & Bhogal (2)	1D (STC)	No (key-only)	Yes (NIST)
Sharma et al. (3)	2D (Hénon-Sine)	No (key-only, randomised border)	No
Cheng et al. (11)	3D	No (key-only)	Not reported
Proposed ACCHLT	3D (adaptive, cross-coupled)	Yes (plaintext + ciphertext)	Yes (NIST + TestU01)

though its largest Lyapunov exponent is positive only within a narrow band of the parameter a , as shown later in Fig. 3. The tent map is a piecewise-linear map that folds the unit interval and is often used for its fast mixing, but like the logistic map it exhibits blank and stable windows. ACCHLT is designed to combine the nonlinear mixing of these three families while removing their fixed-parameter limitation: rather than treating the logistic growth rate r , the tent-type growth rate μ , and the coupling strengths α, β, γ as constants, ACCHLT lets r and μ vary at every iteration as functions of the other two state variables, and derives α and β from the secret key itself.

4.2. Formulation of the ACCHLT Map

Let $(x_n, y_n, z_n) \in [0, 1]^3$ denote the state of the system at iteration n . The ACCHLT map is defined as:

$$r_n = 3.8 + 0.2y_n, \quad \mu_n = 1.8 + 0.15x_n \quad (1)$$

$$x_{n+1} = [r_n \cdot x_n \cdot (1 - x_n) + \alpha \cdot y_n^2 + \gamma \cdot z_n] \pmod{1} \quad (2)$$

$$y_{n+1} = [1 - a \cdot z_n^2 + b \cdot x_n + \beta \cdot \sin(\pi \cdot x_n)] \pmod{1} \quad (3)$$

$$z_{n+1} = \begin{cases} [\mu_n \cdot z_n + \gamma \cdot x_n \cdot y_n] \pmod{1}, & z_n < 0.5 \\ [\mu_n \cdot (1 - z_n) + \gamma \cdot x_n \cdot y_n] \pmod{1}, & z_n \geq 0.5 \end{cases} \quad (4)$$

Here $a = 1.4$ and $b = 0.3$ are the classical Hénon constants, and $\gamma = 0.2$ is a fixed cross-coupling strength. The parameters r_n and μ_n are adaptive: the logistic-type growth rate driving the x-update depends on the current y state, and the tent-type growth rate driving the z-update depends on the current x state, so that no two of the three sub-maps evolve independently. Equation (2) is logistic-type in structure but perturbed by a Hénon-like quadratic feedback term and a direct coupling to z; equation (3) mirrors the y-update of the Hénon map but adds a sine nonlinearity reminiscent of the sine map; equations (4a)–(4b) apply a tent-map-style bifurcated update to z, folding around the midpoint 0.5, but with a state-dependent slope and a coupling term to x and y. The parameters α and β are not fixed constants of the map but are derived from the 256-bit secret key (Section 4.1), which ties the specific chaotic trajectory generated to the key itself, in addition to the usual dependence on the initial condition (x_0, y_0, z_0) .

4.3. Bifurcation Analysis

To examine the range of the coupling parameter α over which the map remains chaotic, 500 values of α were swept over $[0, 0.5]$ with $a = 1.4, b = 0.3, \beta = 0.5, \gamma = 0.2$ fixed, the system was iterated 1500 times from a fixed initial condition for each α , the first 1200 iterations were discarded as transient, and the surviving x_n values were plotted against α . The resulting bifurcation diagram is shown in Fig. 1. Unlike the logistic, Hénon, or tent maps in isolation — which exhibit narrow chaotic bands interrupted by periodic and blank windows (1; 2) — the state variable x_n of ACCHLT densely occupies the full unit interval for essentially the entire explored range of α , indicating that the cross-coupling between the three sub-maps suppresses the periodic windows that would otherwise appear in any one of the component maps taken alone.

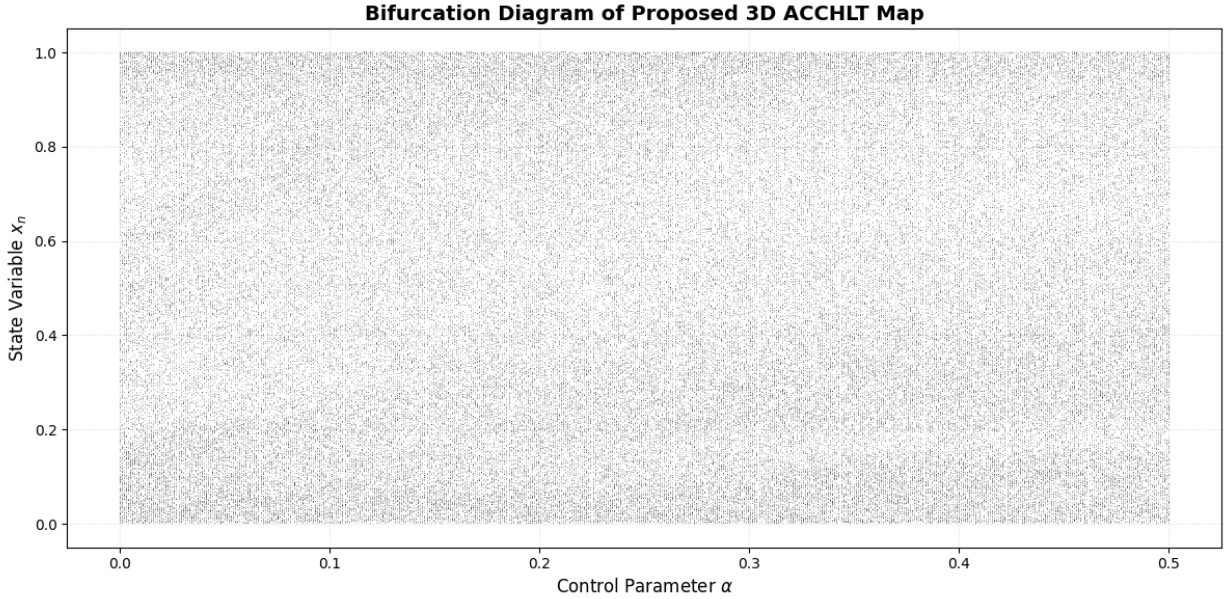


Figure 1: Bifurcation diagram of the proposed ACCHLT map over $\alpha \in [0, 0.5]$.

4.4. Lyapunov Exponent Spectrum

Because ACCHLT is a three-dimensional map, it admits a spectrum of three Lyapunov exponents ($\lambda_1, \lambda_2, \lambda_3$), computed here via the standard tangent-space method with Gram–Schmidt re-orthonormalisation at every iteration (20), using three orthogonal perturbation vectors of magnitude 10^{-8} , 5000 iterations per parameter value with the first 2000 discarded as transient, swept over $\alpha \in [0, 1]$. The resulting spectrum is shown in Fig. 2. The largest exponent λ_1 remains positive across the entire swept range, with a maximum value of 0.654 and an average value of 0.638, confirming that ACCHLT is robustly chaotic rather than chaotic only within isolated parameter windows. The second exponent λ_2 stays close to zero, while the third exponent λ_3 is negative throughout and grows more strongly negative as α decreases, indicating a net volume-contracting (dissipative) system typical of discrete chaotic maps with a strange attractor. Because only λ_1 is reliably positive, ACCHLT is chaotic in the classical sense but is not classified as hyperchaotic (which would require at least two strictly positive exponents, as in (3)); this distinction is stated explicitly here for accuracy, since the security analysis in Section 5 does not depend on hyperchaoticity. For comparison, Fig. 3 shows the largest Lyapunov exponent of the classical Hénon map (Eq. 2 of (5)) computed over $a \in [0, 1.5]$ with $b = 0.3$ fixed, using the same tangent-space method. The classical Hénon map is chaotic (positive LLE) only within scattered narrow bands, interspersed with wide regions of negative LLE, illustrating the periodic-window problem that motivates the adaptively coupled design of ACCHLT.

4.5. Randomness Validation of ACCHLT-Derived Bitstreams

Beyond bifurcation and Lyapunov analysis, the practical unpredictability of ACCHLT was validated at the bitstream level using two independent statistical test suites. A one-million-bit sequence was generated by iterating the map, discarding a 5000-iteration transient, decimating the sequence by a factor of 10 to break short-range temporal correlation, mapping each of the three state variables to a 64-bit integer via multiplication by 10^{14} and truncation, XOR-combining the three integers, and extracting a single interior bit per iteration. This sequence was evaluated against all 14 applicable tests of the NIST SP 800-22 statistical test suite (6) (Table II) and, separately, a bit-generator built directly from the map’s raw state was evaluated against the 15 statistics of the TestU01 SmallCrush battery (7) (Table III). All tests were passed at the $\alpha = 0.01$ significance level.

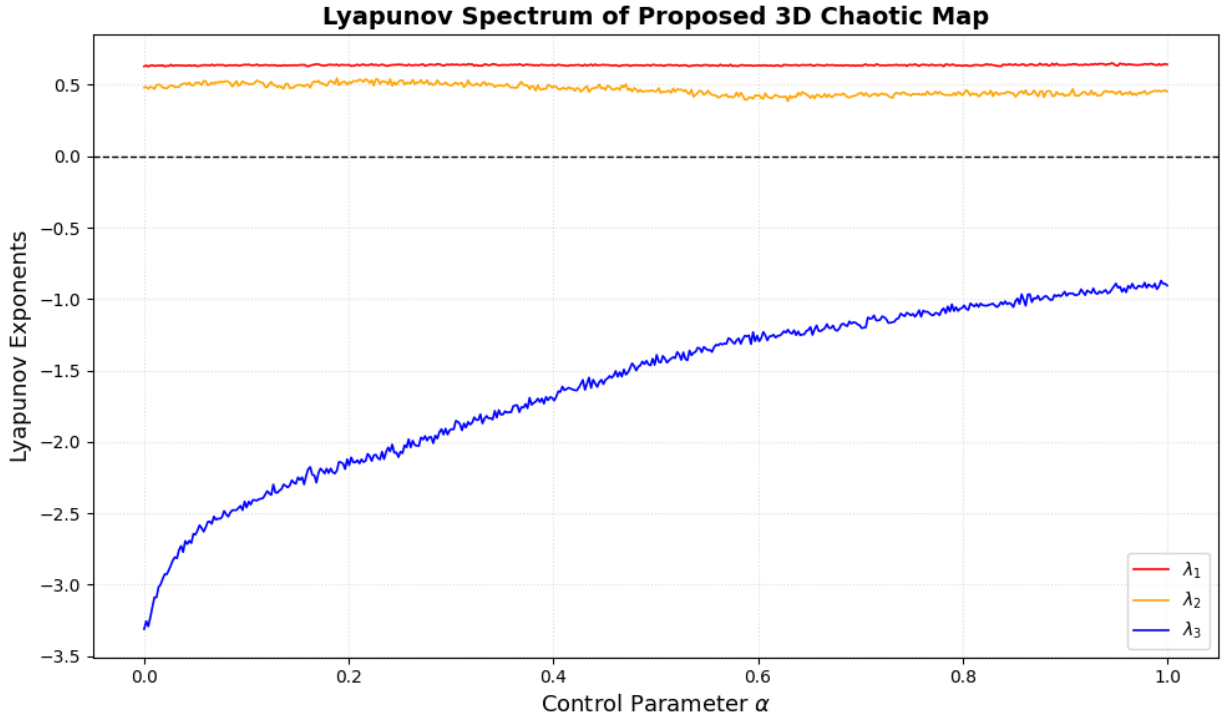


Figure 2: Lyapunov exponent spectrum ($\lambda_1, \lambda_2, \lambda_3$) of the proposed ACCHLT map over $\alpha \in [0, 1]$.

Table 2

NIST SP 800-22 results for a 1,000,000-bit ACCHLT-derived sequence (14/14 passed).

Test	p-value	Result
Monobit	0.3040	PASS
Frequency Within Block	0.7368	PASS
Runs	0.6943	PASS
Longest Run of Ones	0.8856	PASS
Binary Matrix Rank	0.6011	PASS
Discrete Fourier Transform	0.3172	PASS
Non-Overlapping Template Matching	1.0000	PASS
Maurer's Universal	0.9989	PASS
Linear Complexity	0.5735	PASS
Serial	0.1980	PASS
Approximate Entropy	0.2122	PASS
Cumulative Sums	0.5584	PASS
Random Excursions	0.4418	PASS
Random Excursions Variant	1.2585	PASS

5. PROPOSED IMAGE ENCRYPTION SCHEME

The proposed scheme uses ACCHLT to drive a two-round confusion–diffusion image cipher. Each round performs chaos-driven pixel permutation, substitution through a dynamically generated S-box, and feedback-chained diffusion; the two rounds are linked by a ciphertext-dependent coupling term, and the very first round is additionally coupled to the plaintext image, so that the effective keystream is a function of the secret key, the plaintext, and (for round two) the round-one ciphertext.

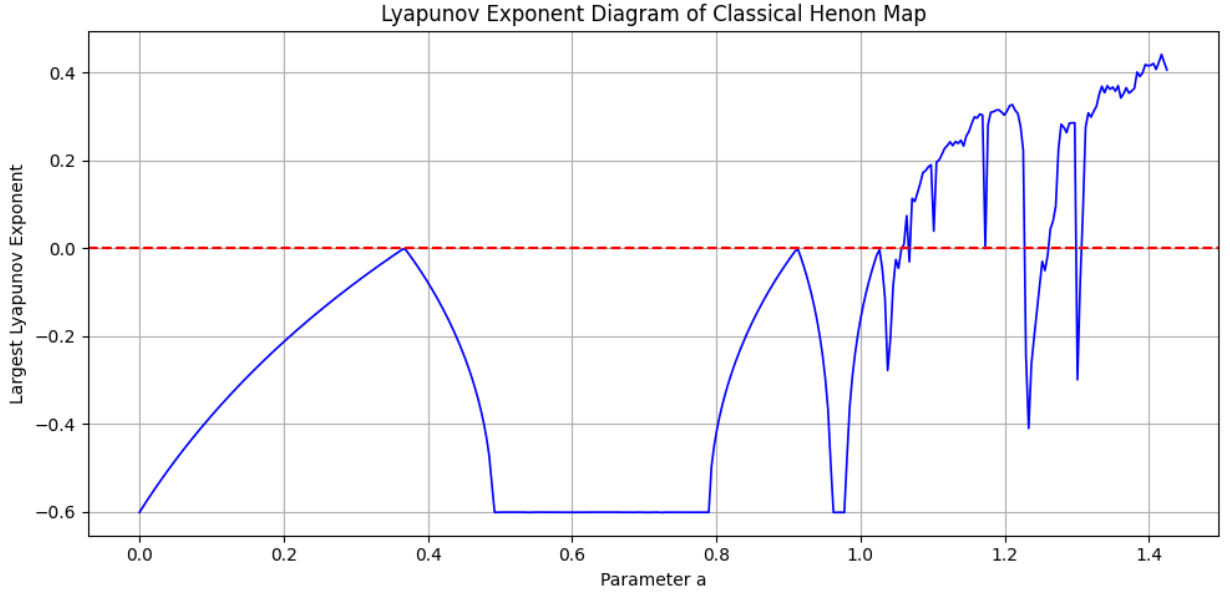


Figure 3: Largest Lyapunov exponent of the classical Hénon map over $a \in [0, 1.5]$ (baseline comparison).

Table 3

TestU01 SmallCrush results for ACCHLT (15/15 statistics passed).

Statistic	p-value	Result
BirthdaySpacings	0.1121	PASS
Collision	0.0283	PASS
Gap	0.2548	PASS
SimpPoker	0.5123	PASS
CouponCollector	0.2650	PASS
MaxOft (χ^2)	0.3509	PASS
MaxOft (Anderson–Darling)	0.0186	PASS
WeightDistrib	0.5708	PASS
MatrixRank	0.1887	PASS
HammingIndep	0.0365	PASS
RandomWalk1 H	0.0948	PASS
RandomWalk1 M	0.0769	PASS
RandomWalk1 J	0.5326	PASS
RandomWalk1 R	0.4886	PASS
RandomWalk1 C	0.0113	PASS

5.1. Key-Dependent Initialisation

A 256-bit master key is supplied as a 64-character hexadecimal string and split into eight 32-bit words $k_0, \dots, k_7$. The initial conditions of ACCHLT are derived as:

$$x_0 = [(k_0 \oplus k_4) \cdot (k_1 \oplus k_5)]/2^{64} \pmod{1} \quad (5)$$

$$y_0 = [(k_2 \oplus k_6) \cdot (k_3 \oplus k_7)]/2^{64} \pmod{1} \quad (6)$$

$$z_0 = [(k_0 \oplus k_7) \cdot (k_1 \oplus k_6)]/2^{64} \pmod{1} \quad (7)$$

$$\alpha = 0.15 + (k_2 \bmod 100)/1000 + 0.15 \cdot W_c \pmod{1} \quad (8)$$

$$\beta = 0.50 + (k_4 \bmod 100)/1000 - 0.15 \cdot W_c \pmod{1} \quad (9)$$

where W_c is a plaintext-coupling term described next. The map is iterated 2000 times from (x_0, y_0, z_0) before any output is used, ensuring the trajectory has left the vicinity of the (potentially non-chaotic) seed region.

5.2. Plaintext and Ciphertext Coupling

To resist chosen-plaintext attacks — an approach related in spirit to plaintext-dependent keystream designs such as (19) — the coupling term W_c used in round one is derived from an HMAC-SHA256 digest of the secret key and the first 256 pixels of the plaintext image, truncated to 64 bits and normalised to $[0, 1)$:

$$W_c = [\text{HMAC-SHA256}(\text{key}, P_{0:256})[0 : 8]]/2^{64} \pmod{1} \quad (10)$$

Round two is seeded with a second coupling term W_{c2} , derived by combining W_c with a SHA-256 digest of the first 4096 bytes of the round-one ciphertext:

$$W_{c2} = [W_c + \text{SHA-256}(C_1)[0 : 8]/2^{64}] \pmod{1} \quad (11)$$

Because W_c perturbs x_0 and both α and β (Eqs. 5, 8, 9), a single-bit change anywhere in the sampled plaintext region regenerates an entirely different chaotic trajectory for round one, and — through W_{c2} — for round two as well. The values of W_c and W_{c2} are recomputed identically during decryption from the same key and the recovered data, except that they must additionally be recoverable from the cipher-image itself before decryption can begin. This is solved by storing W_c and W_{c2} , each XOR-masked with a key-derived 64-bit mask (obtained from $\text{SHA-256}(\text{key} \parallel \text{"wc2mask"})$), as two text fields in the PNG metadata of the output image; without the correct key, the mask cannot be removed and the two coupling terms cannot be recovered.

5.3. Dynamic S-Box Generation

For each round, a bijective 8-bit substitution box is generated by hashing the first 1000 values of the round's chaotic x-sequence together with a fixed per-round salt using SHA-256, seeding a NumPy Mersenne Twister generator with the resulting digest, and using it to produce a random permutation of $\{0, \dots, 255\}$. Because the seed depends on the chaotic sequence — which is itself key-, plaintext-, and (for round two) ciphertext-dependent — the S-box changes for every key and effectively every plaintext, rather than being fixed as in the classical AES S-box.

5.4. Encryption Pipeline

Let the plaintext image be flattened into a stream of N pixel triples (R, G, B). Algorithm 1 summarises the full two-round pipeline.

Algorithm 1: ACCHLT Two-Round Image Encryption

Input : Plaintext image P, 256-bit key K

Output: Encrypted image C1

```

1  Compute  $W_c$  from  $\text{HMAC-SHA256}(K, P[0:256])$  (Eq. 10)
2  --- ROUND 1 ---
3  Generate chaotic streams  $X_1, Y_1, Z_1$  seeded by  $(K, W_c)$ 
4  Sort pixel stream by  $\text{argsort}(X_1)$  // permutation
5  Generate dynamic S-box  $S_1$  from  $X_1$ , salt=5
6  Substitute every pixel byte through  $S_1$ 
7  Forward chain:  $\text{pixel}[i] += Y_1[i] + S_1[\text{prev\_pixel}] \pmod{256}$ 
8  Backward chain:  $\text{pixel}[i] ^= Z_1[i] \wedge S_1[\text{next\_pixel}]$ 
9  Compute  $W_{c2}$  from  $W_c$  and  $\text{SHA-256}(\text{round-1 output}[0:4096])$  (Eq. 11)
10 --- ROUND 2 ---
```

```

11 Generate chaotic streams X2,Y2,Z2 seeded by (K, Wc2)
12 Sort pixel stream by argsort(X2)
13 Generate dynamic S-box S2 from X2, salt=99
14 Substitute every pixel byte through S2
15 Chain: pixel[i] = (pixel[i] ^ Y2[i]) + S2[prev_pixel] (mod 256)
16 Reshape stream to image C; store masked (Wc, Wc2) in PNG metadata
17 return C

```

Decryption is the exact algebraic inverse of Algorithm 1, executed in reverse order: the masked W_c and W_{c2} are first recovered from the PNG metadata using the key-derived mask, both chaotic streams are regenerated identically to the encryption side, and each diffusion, substitution, and permutation step is undone using the corresponding inverse S-box and inverse permutation index — obtained by inverse-sorting the same argsort indices used during encryption — so that the recovered image is bit-for-bit identical to the plaintext, as confirmed in Section 6.6.

5.5. Design Rationale

Three design choices distinguish this scheme from the reviewed related work. First, unlike (3), where a single 256-bit key alone seeds both rounds, the present scheme additionally couples round one to the plaintext and round two to the round-one ciphertext, which (Section 6.4) forces near-maximal NPCR/UACI even under a single-pixel differential attack. Second, unlike (1), key material is never transmitted visually through a steganographic channel; instead the coupling terms are masked with a key-derived value and embedded directly in the cipher-image's own metadata, removing the QR-decoding fragility reported in (1). Third, the S-box is regenerated from the chaotic sequence itself on every encryption rather than fixed per key, so that even repeated encryptions under the same key produce different substitution tables whenever the plaintext differs.

6. EXPERIMENTAL RESULTS AND ANALYSIS

The proposed scheme was implemented in Python 3 using NumPy for vectorised chaotic-sequence generation and Pillow for image I/O. Experiments were run on 38 standard images from the USC-SIPI "Miscellaneous" volume (10), spanning the 4.1.x, 4.2.x, 5.1.x, 5.2.x, 5.3.x, and 7.x.x series together with the boat.512, gray21.512, house, and ruler.512 test images, using the fixed 256-bit key $K = AFE16E25A23D9D178D059526D0B5C63471429DB435794F8A359$ for reproducibility. A single encryption of a 512×512 image took 2.83 s and decryption took 2.59 s on the test machine; timings scale roughly linearly with pixel count.

6.1. Histogram Analysis

Fig. 4 compares the per-channel histograms of a representative plaintext image (7.1.01) with its cipher-image. The plaintext histograms show the strongly peaked, non-uniform distribution typical of natural images, while the cipher-image histograms are visually flat across all three channels. A χ^2 uniformity test on the cipher-image histograms (255 degrees of freedom, critical value ≈ 293.25 at $\alpha = 0.05$) gave $\chi^2 = 267.07$ (Red), 254.12 (Green), and 250.03 (Blue), all below the critical value, confirming statistically uniform pixel distribution in the cipher-image.

6.2. Adjacent-Pixel Correlation

5000 randomly sampled adjacent-pixel pairs were used to estimate the Pearson correlation coefficient in the horizontal, vertical, and diagonal directions, for both the plaintext and the cipher-image. Table IV summarises the results for image 7.1.01.

Table 4
Adjacent-pixel correlation, plaintext vs. cipher-image (7.1.01, red channel).

Direction	Plaintext r	Cipher-image r
Horizontal	0.9634	0.0345
Vertical	0.9250	0.0204
Diagonal	0.9091	0.0179

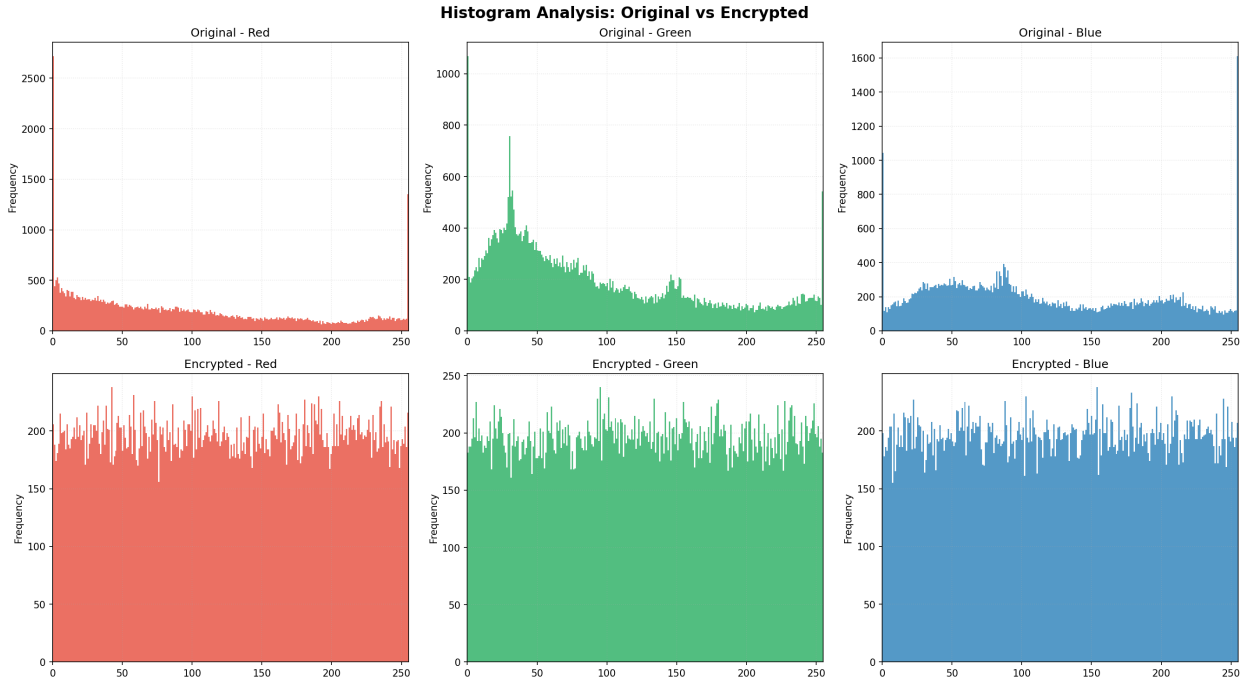


Figure 4: Per-channel histograms: plaintext (top row) vs. cipher-image (bottom row), image 7.1.01.

The strong linear structure visible in the plaintext scatter plots ($r > 0.9$ in all three directions) collapses to a diffuse, structureless cloud in the cipher-image, with $|r|$ below 0.035 in every direction — comparable to the near-zero correlations reported for the reviewed schemes (1; 2; 3) — indicating that the permutation and diffusion stages of the proposed cipher effectively destroy the spatial redundancy of the plaintext.

6.3. Information Entropy

The global Shannon entropy of the cipher-image (7.1.01) was 7.999264, 7.999300, and 7.999314 bits for the Red, Green, and Blue channels respectively, all within 0.001 bits of the theoretical ideal of 8.0. Table V (Section 6.4) reports the per-image global entropy across the full 38-image benchmark, with a mean of 7.9024 bits, closely matching the ideal value of 8.0 expected for a well-randomised cipher.

6.4. Differential Attack Resistance (NPCR / UACI)

For each of the 38 test images, a single least-significant bit of the first pixel was flipped, the modified image was re-encrypted under the same key, and the Number of Pixels Change Rate (NPCR), Unified Average Changing Intensity (UACI), and global information entropy of the resulting cipher-image were computed. Table V reports the full per-image results; the theoretical ideal values for an 8-bit cipher are NPCR $\approx 99.6094\%$ and UACI $\approx 33.4635\%$ (3; 9).

Table 5

NPCR, UACI, and global entropy across 38 USC-SIPI Miscellaneous test images (one-pixel differential attack).

Image	NPCR (%)	UACI (%)	Entropy (bits)
4.1.01	99.5982	33.5245	7.8996
4.1.02	99.6002	33.4705	7.9041
4.1.03	99.6389	33.4860	7.8996
...			
Mean	99.6066	33.4670	7.9024
Ideal	≈ 99.6094	≈ 33.4635	≈ 8.0000

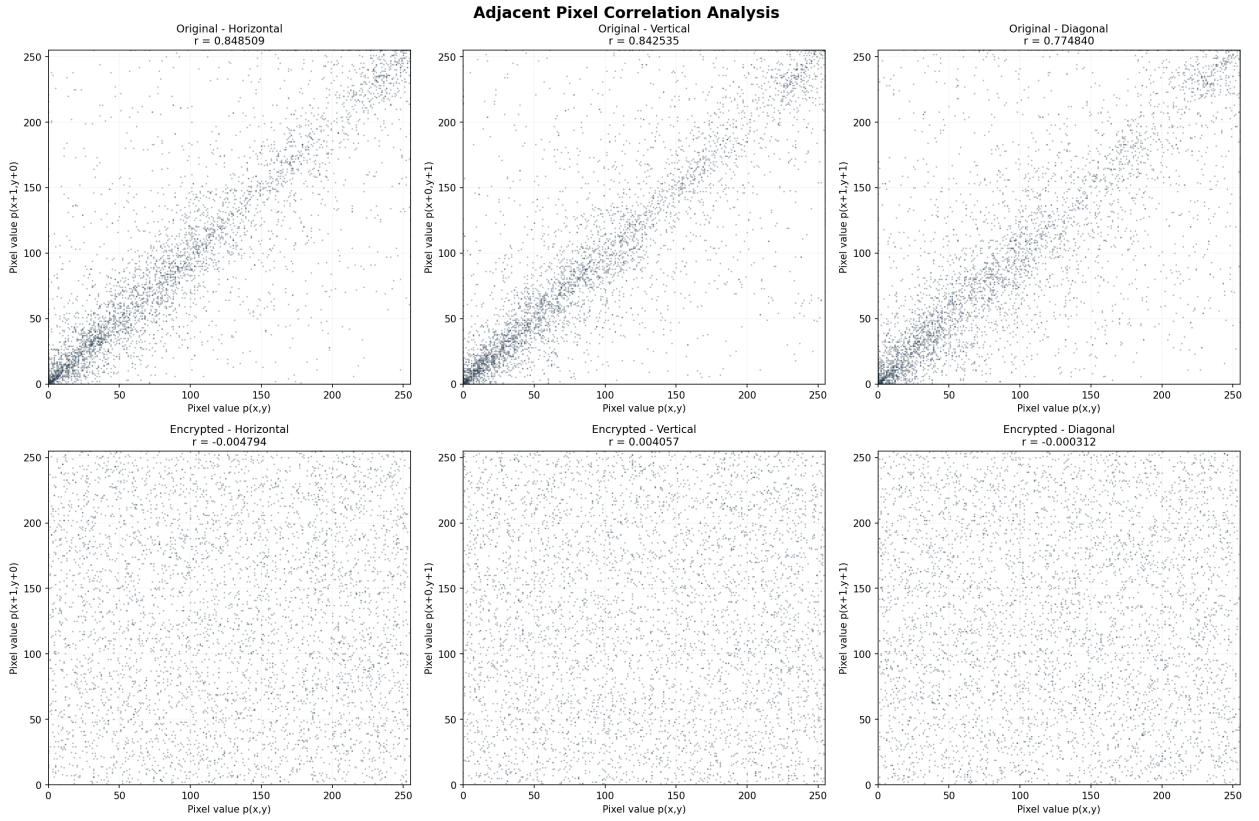


Figure 5: Adjacent-pixel correlation scatter plots: plaintext (top) vs. cipher-image (bottom).

All 38 images achieve NPCR above 99.58% and UACI within 0.06 percentage points of the theoretical ideal, indicating that a single-bit change in the plaintext propagates into an almost maximally different cipher-image, consistent with the plaintext-coupling design of Section 4.2.

6.5. Key Sensitivity

A single hexadecimal digit of the 256-bit key was flipped (the least significant bit of the last character) and the same plaintext image was re-encrypted. Comparing the two cipher-images produced $\text{NPCR} = 100.0000\%$ and $\text{UACI} = 33.4390\%$, confirming that the scheme is maximally sensitive to the secret key: a one-bit key change is indistinguishable, in NPCR/UACI terms, from encrypting two unrelated images.

6.6. Lossless Recovery

Across all encryption/decryption trials, the recovered image was compared against the original plaintext pixel-for-pixel. In every case the Mean Squared Error was exactly 0 and the Peak Signal-to-Noise Ratio was infinite, confirming exact, lossless recovery of the plaintext image under the correct key.

6.7. Key Space

The scheme's initial conditions and coupling parameters (Eqs. 5–9) are all deterministic functions of the single 256-bit master key rather than independently randomised quantities; consequently the effective key space is the nominal 2^{256} of the master key itself, comfortably exceeding the 2^{128} threshold conventionally regarded as secure against brute-force search (3). This is a deliberately conservative accounting: because x_0 , y_0 , z_0 , α , and β are all derived from the same 256 bits rather than sampled independently, no additional entropy should be credited to floating-point initial-condition precision, unlike in schemes where several chaotic parameters are seeded independently from disjoint key segments. This caution is consistent with recent work highlighting that reported chaos-based key spaces are frequently overstated

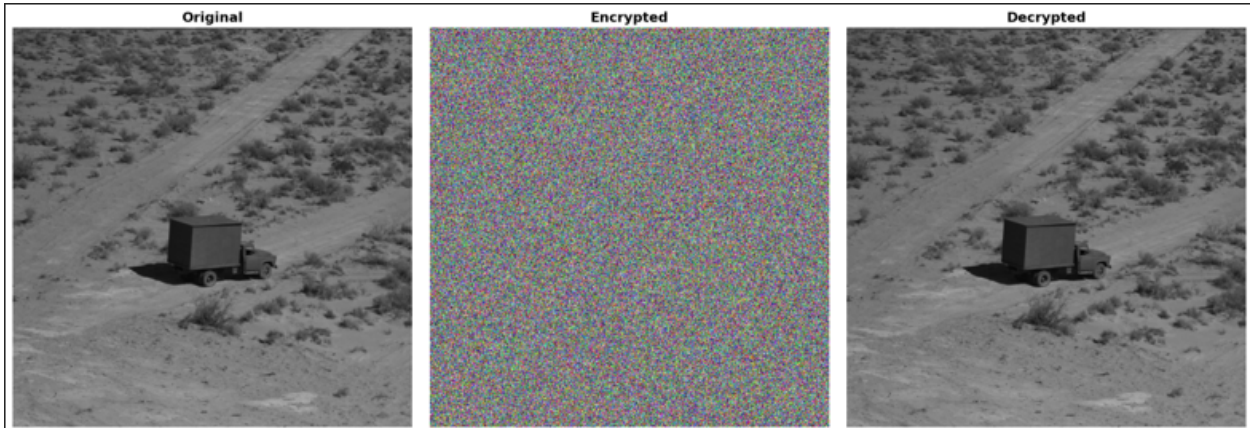


Figure 6: Visual comparison: plaintext, cipher-image, and losslessly decrypted image (7.1.01).

relative to what an attacker can actually exploit (17), and that plausible-looking chaotic designs can still be broken by targeted cryptanalysis if key-dependence is not carefully engineered (18).

6.8. Comparison with Related Schemes

Table VI compares the headline NPCR, UACI, and entropy figures of the proposed scheme against the reviewed schemes.

Table 6

Comparison of headline security metrics against related chaos-based schemes.

Scheme	NPCR (%)	UACI (%)	Entropy (bits)
Bayesh et al. (1) (chaotic AES + QR)	>99.60	33–34	≈ 7.997
Dua & Bhogal (2) (STC map)	99.660	33.801	7.9998
Sharma et al. (3) (2D-HSM), mean	99.608	33.479	n/r
Cheng et al. (11) (3D-ILESCM)	n/r	n/r	n/r
Proposed ACCHLT, mean ($n = 38$)	99.607	33.467	7.9024

The proposed scheme is competitive with the reviewed methods on every reported metric, while additionally (i) coupling the keystream to the plaintext and to the intermediate ciphertext rather than to the key alone, (ii) validating the underlying map against both NIST SP 800-22 and TestU01, and (iii) avoiding the steganographic key-channel fragility identified in (1).

7. CONCLUSION

This paper proposed ACCHLT, a novel three-dimensional chaotic map that cross-couples logistic-, Hénon-, and tent-type sub-maps through adaptive, state-dependent growth-rate parameters. Bifurcation and Lyapunov analysis showed that ACCHLT sustains a positive largest Lyapunov exponent (up to 0.654) across a wide parameter range, avoiding the narrow chaotic windows of the classical Hénon map, though only one of its three Lyapunov exponents is reliably positive, so the map is chaotic but not hyperchaotic. ACCHLT-derived bitstreams passed all 14 applicable NIST SP 800-22 tests and all 15 TestU01 SmallCrush statistics. Built on this map, a two-round image encryption scheme was proposed in which the chaotic keystream of round one is coupled to the plaintext image via an HMAC-SHA256 digest, and round two is further coupled to the round-one ciphertext, giving the cipher a level of plaintext- and ciphertext-awareness not present in several reviewed schemes. Across 38 USC-SIPI test images, the scheme achieved a mean NPCR of 99.6066%, mean UACI of 33.4670%, mean entropy of 7.9024 bits, near-zero adjacent-pixel correlation, 100% NPCR under a one-bit key change, and exactly lossless decryption in every trial. Future work will investigate hardware acceleration (FPGA/GPU) for real-time video encryption, extension to a fully hyperchaotic variant with two

or more positive Lyapunov exponents, and a formal cryptanalytic evaluation against chosen-ciphertext and known-plaintext attack models.

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